

THE AFFINE BERNSTEIN THEOREM IN DIMENSIONS THREE THROUGH NINE

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ABSTRACT. We prove the Euclidean-complete affine Bernstein conjecture in dimensions three through nine: a smooth locally uniformly convex affine maximal graph is an elliptic paraboloid whenever its induced Euclidean metric is complete. The same conclusion holds, without an initial graph assumption, for connected smooth open (noncompact and without boundary) affine-complete locally uniformly convex immersed hypersurfaces that are classically affine maximal in these dimensions.

1. INTRODUCTION

The affine Bernstein problem asks whether a complete locally uniformly convex affine maximal hypersurface must be an elliptic paraboloid. The meaning of completeness is essential: completeness for the metric induced from Euclidean space and completeness for the affine metric are different hypotheses. We prove the Euclidean-complete graph assertion in dimensions three through nine.

Let $\Omega \subset \mathbb{R}^n$ be a nonempty open convex domain, and let $u \in C^\infty(\Omega)$ have positive-definite Hessian. Write

$$U^{ij} = \det(D^2u)(D^2u)_{ij}^{-1}, \quad w = (\det D^2u)^{-(n+1)/(n+2)}.$$

The affine maximal equation is

$$(1.1) \quad U^{ij}w_{ij} = 0.$$

It is the Euler equation for affine area $\int (\det D^2u)^{1/(n+2)} \, dx$.

Theorem 1.1. *Let $3 \leq n \leq 9$. Suppose $u \in C^\infty(\Omega)$ has positive-definite Hessian and satisfies (1.1). If its graph is complete for the induced Euclidean metric*

$$g_{ij} = \delta_{ij} + u_i u_j,$$

then $\Omega = \mathbb{R}^n$ and

$$u(x) = \frac{1}{2}x^\top A x + b \cdot x + c$$

for a positive-definite symmetric matrix A , a vector b , and a constant c . In particular, the graph is an invertible affine image of the standard elliptic paraboloid.

Theorem 1.1 resolves the Euclidean-complete affine Bernstein conjecture for smooth locally uniformly convex graphs in dimensions three through nine. It includes the entire graph assertion and imposes neither a growth bound nor completeness of the affine metric.

Chern asked whether an entire locally uniformly convex affine maximal surface must be an elliptic paraboloid [5]. Calabi developed the affine variational framework and proved the surface conclusion under both Euclidean and affine completeness [4]; see also [10, Section 1]. For a graph, the affine (Berwald–Blaschke) metric is

$$g_{ij}^{\text{aff}} = (\det D^2u)^{-1/(n+2)} u_{ij},$$

so its completeness is a different condition from that in Theorem 1.1. Trudinger and Wang proved the Euclidean-complete surface theorem, dispensing with affine completeness [9, Theorem 1.1]. Li and Jia gave a separate proof of the surface theorem under affine completeness alone [7].

The analytic reduction in higher dimensions is also due to Trudinger and Wang. Their interior estimates and rescaling argument prove rigidity for Euclidean-complete graphs in every dimension provided the separation of the graph from its tangent planes has a modulus of strict convexity that remains uniform under normalization of tangent-plane sublevel sets [9, Theorem 4.2 and Corollary 4.3]. They establish this uniform control in dimension two by studying convex limits and their supporting contact sets [9, Section 5]. Obtaining it in higher dimensions without an additional hypothesis is the obstacle addressed here. Later work has developed alternative analytic approaches: Wang and Zhou give a proof of the entire-surface theorem using the partial Legendre transform, and higher-dimensional rigidity results under integral bounds on the Hessian and the inverse Hessian determinant [12, Theorem 1.2 and Corollary 1.6]. Our proof instead derives the required convexity control from the geometry of affine limits, with no growth or integral bound assumed.

For locally uniformly convex hypersurfaces in dimension at least two, Trudinger and Wang proved that affine completeness implies Euclidean completeness [10, Theorem A]; the converse fails in general. Their global graph theorem [10, Corollary 1] therefore extends our conclusion to connected smooth open affine-complete locally uniformly convex immersed hypersurfaces that are classically affine maximal, without an initial graph assumption. The precise statement and proof appear in Corollary 6.5.

The singular entire affine maximal graph constructed by Trudinger and Wang in dimension ten [9, Section 7] is a weak solution. It is not smooth and does not furnish a counterexample under the hypotheses of Theorem 1.1. Our argument identifies the dimension restriction in a quadratic coercivity estimate, whose strict positivity holds through dimension nine.

The distinction between the classical equation and its determinant-power generalizations is also relevant. For affine maximal type equations with $w = (\det D^2 u)^a$, Sun, Xing and Xu construct nonquadratic Euclidean-complete examples in every dimension $n \geq 2$ for $a \in [-n/(n+1), 0)$ [8, Theorem 1.3]. This interval excludes the classical exponent $-(n+1)/(n+2)$ in Equation (1.1).

The main ideas. For $a \in \Omega$, let

$$l_a(x) = u(a) + Du(a) \cdot (x - a), \quad S_u(a, t) = \{x \in \Omega : u(x) < l_a(x) + t\} \quad (t > 0).$$

These tangent sections measure the separation of the graph from its supporting planes. The geometric objective is uniform balance: for one $\rho > 0$, reflection about the base point followed by contraction by ρ takes every closed section into itself. This control must hold at every base point and every height.

We obtain balance by studying full-dimensional closed convex limits of affine images of the epigraph $D = \{(x, z) : x \in \Omega, z \geq u(x)\}$. Write the coordinates of such a limit as $(s, y) \in \mathbb{R}^k \times \mathbb{R}^{n+1-k}$. A model with k directions is supported in $s_i \geq 0$, contains $\{(s, 0) : s_i \geq 0\}$, and has compact caps cut out by $\sum_i s_i \leq T$. Its transverse fibers are $K_s = \{y : (s, y) \in D_{\text{lim}}\}$, where D_{lim} denotes the limiting body. Maximizing k forces a uniform centering estimate $-K_s \subset C_* K_s$ for all positive fibers: failure of this estimate produces an additional nonnegative direction by a further affine rescaling. Normalized caps and convex limits also play a central role in the two-dimensional argument of Trudinger and Wang [9, Section 5]; here the rescaling is organized by the maximal number of nonnegative directions.

The analytic step excludes $k \geq 2$. Support functions of the fibers give a positive second form and a natural invariant measure. The affine maximal equation yields an integral inequality in logarithmic coordinates, forcing exponential growth of a related measure. A cap identity and estimates for Hessian minors bound its mass on each normalized log cell, allowing only polynomial

growth. Both estimates are used on smooth approximants over one sufficiently large finite box; the limiting body needs no regularity.

These two ingredients—increasing the number of model directions when centering degenerates, and combining a logarithmic integral inequality with affine-invariant cell bounds—are the principal technical contributions. The resulting uniform balance of tangent sections supplies a scale-independent modulus of strict convexity. We then use the interior estimates and section-rescaling argument of Trudinger and Wang [9, Sections 2 and 4]: normalized third derivatives stay bounded, while transforming them back gives a bound tending to zero. John’s ellipsoid theorem [6, 1] supplies the normalizations. The compactness arguments and the cap and tube estimates used here are proved below.

Notation and organization. Closed sections use the corresponding non-strict inequality. The symbol $B_r(x)$ denotes an open Euclidean ball, and $B_r = B_r(0)$; its ambient dimension will be clear. The notation H^{-1} for a positive second form specifies the dual norm on covectors. All constants in uniform estimates may depend on the fixed dimension and, where stated, on the family of affine limits of the given epigraph.

Section 2 establishes the compactness and centering argument. Sections 3 and 4 derive the cap identity and the tube inequality. Section 5 excludes models with several directions, and Section 6 deduces section control and proves Theorem 1.1.

2. CONVEX LIMITS AND CENTERED FIBERS

The objective of this section is to show that fibers are uniformly centered once the number of model directions is maximal. We first obtain compact caps from graph completeness, then normalize a hypothetical sequence of increasingly asymmetric fibers. Its limit supplies the additional direction that contradicts maximality.

Throughout this section, $n \geq 2$, $\Omega \subset \mathbb{R}^n$ is nonempty, open and convex, and $u \in C^\infty(\Omega)$ has positive-definite Hessian. We assume only that its graph is complete for the induced Euclidean metric. No differential equation is needed in this section. For $a \in \Omega$, write

$$l_a(x) = u(a) + Du(a) \cdot (x - a), \quad g_a(x) = u(x) - l_a(x).$$

Lemma 2.1. *At every finite boundary point $b \in \partial\Omega$,*

$$\lim_{\Omega \ni x \rightarrow b} u(x) = +\infty.$$

The epigraph

$$D = \{(x, z) : x \in \Omega, z \geq u(x)\}$$

is a closed full-dimensional convex set with boundary equal to the graph of u . For every $a \in \Omega$ and $t \geq 0$, the set $\{x \in \Omega : g_a(x) \leq t\}$ is compact and contained in Ω . Consequently the cap $D \cap \{z - l_a(x) \leq t\}$ is compact.

Proof. Suppose $x_j \rightarrow b \in \partial\Omega$ and $u(x_j) \leq A$. Fix $a \in \Omega$. Convexity, followed by continuity at the interior point $a + q(b - a)$, gives

$$u(a + q(b - a)) \leq (1 - q)u(a) + qA, \quad 0 \leq q < 1.$$

An affine supporting function at a gives a lower bound on the same segment. The resulting bounded convex function of q has finite total variation and a finite limit as $q \uparrow 1$. Its lifted curve has finite Euclidean length, since

$$\int_0^1 \sqrt{|b - a|^2 + \left| \frac{d}{dq} u(a + q(b - a)) \right|^2} dq \leq |b - a| + \int_0^1 \left| \frac{d}{dq} u(a + q(b - a)) \right| dq < \infty.$$

Its tail is therefore Cauchy in the graph metric, but its base coordinate tends to $b \notin \Omega$. This contradicts completeness. The asserted boundary limit follows. It also shows that a bounded convergent sequence of points of D cannot have its base limit on $\partial\Omega$. Thus D is closed, and its boundary is exactly the graph.

Choose $r, c > 0$ such that $\overline{B}_r(a) \subset \Omega$ and $D^2u \geq c\text{Id}$ on this ball. Along every ray from a , convexity of g_a , together with $g_a(a) = 0$, gives

$$g_a(x) \geq \frac{cr}{2}|x - a| \quad (x \in \Omega, |x - a| \geq r).$$

Indeed, $g_a \geq cr^2/2$ at the point of the ray at distance r , and its quotient by the distance is nondecreasing. Thus each stated sublevel is bounded. The boundary limit just proved makes it closed in $\mathbb{R}^n$ and contained in Ω . On such a compact base, both l_a and the cap height are bounded, which proves the last claim. $\square$

We use local convergence of nonempty closed convex sets in the sense of locally uniform convergence of their distance functions. This topology has the following elementary properties. Every point of a limit is approximated by points of the approximating sets, and every limit of a convergent sequence of such points belongs to the limit. If the limit has nonempty interior, every compact subset of its interior is eventually contained in the interiors of the approximants. For the latter statement, enclose a point in a simplex whose vertices lie in the limit, approximate its vertices, and then use a finite cover of the compact subset. A sequence of closed convex sets meeting a fixed bounded set has a locally convergent subsequence: their distance functions are locally equibounded and 1-Lipschitz, and a diagonal application of compactness gives the distance function of a nonempty closed convex set.

Let $\mathcal{F}(D)$ be the family of all full-dimensional local limits of invertible affine images of D , including those images themselves. The family is invariant under invertible affine maps and closed under further full-dimensional local limits. To justify the last assertion explicitly, the distance-function topology is metrizable by

$$d_{\text{loc}}(C, E) = \sum_{q=1}^{\infty} 2^{-q} \min \left\{ 1, \sup_{|X| \leq q} |\text{dist}(X, C) - \text{dist}(X, E)| \right\}.$$

If $C_j \in \mathcal{F}(D)$ tends to a full-dimensional set C , choose an affine image E_j of D with $d_{\text{loc}}(E_j, C_j) < 1/j$. Then $E_j \rightarrow C$. Affine invariance follows because a fixed invertible affine map and its inverse take bounded sets to bounded sets and preserve the two pointwise convergence properties above. Thus affine maps varying with j may be applied before this diagonal choice.

Lemma 2.2. *Suppose $C_j \rightarrow C$ locally, and let ℓ be an affine function. If $C \cap \{\ell \leq b\}$ is bounded and contains a point q with $\ell(q) < b$, then the caps $C_j \cap \{\ell \leq b\}$ are eventually uniformly bounded. They converge in Hausdorff distance to $C \cap \{\ell \leq b\}$.*

Proof. Choose $q_j \in C_j$ with $q_j \rightarrow q$, so $\ell(q_j) < b$ eventually. If points $p_j \in C_j \cap \{\ell \leq b\}$ escape every bounded set, pass to a subsequence for which

$$\frac{p_j - q_j}{|p_j - q_j|} \longrightarrow v, \quad |v| = 1.$$

For each fixed $t \geq 0$, the points $q_j + t(p_j - q_j)/|p_j - q_j|$ eventually lie on the segments $[q_j, p_j]$. Their limit $q + tv$ therefore belongs to $C \cap \{\ell \leq b\}$, contradicting boundedness.

Uniform boundedness and local convergence show that every limit point of the approximating caps lies in the limit cap. Conversely, if x belongs to the limit cap, then $(1 - \eta)x + \eta q$ lies strictly below the top for every $0 < \eta \leq 1$ and is approximated by points of the approximating caps. Letting $\eta \downarrow 0$ and using compactness gives the reverse Hausdorff inclusion. $\square$

Definition 2.3. A *model with k directions*, or a *k -model*, is a member C of $\mathcal{F}(D)$ equipped with affine coordinates

$$(s, y) \in \mathbb{R}^k \times \mathbb{R}^m, \quad m = n + 1 - k, \quad 1 \leq k \leq n + 1,$$

such that

$$\{(s, 0) : s_i \geq 0\} \subset C \subset \{(s, y) : s_i \geq 0\},$$

and every cap $C \cap \{\sum_i s_i \leq T\}$, $T \geq 0$, is compact. Its transverse fibers are

$$K_s = \{y : (s, y) \in C\}.$$

The coordinates are part of the model data.

Lemma 2.4. *For a k -model, addition of any vector $(v, 0)$ with $v_i \geq 0$ preserves C . Every fiber K_s with $s_i > 0$ is compact, full-dimensional in $\mathbb{R}^m$, and contains zero. Moreover,*

$$(2.1) \quad K_s \subset K_t \quad (s_i \leq t_i), \quad K_{\lambda s} \subset \lambda K_s \quad (\lambda \geq 1).$$

At least one 1-model exists.

Proof. The recession assertion holds for any closed convex set containing the indicated orthant, without a full-dimensionality assumption. For $X \in C$ and $v_i \geq 0$, take the limit as $\eta \downarrow 0$ of

$$(1 - \eta)X + \eta(v/\eta, 0) \in C$$

to obtain $X + (v, 0) \in C$. This proves fiber monotonicity. Since the ambient origin belongs to C , contraction toward it proves the second inclusion in (2.1). Containment of zero and compactness follow from the definition.

Take an interior ball of C centered at (s^0, y^0) ; all coordinates of s^0 are positive. Given $s_i > 0$, choose $0 < \eta \leq 1$ with $\eta s_i^0 \leq s_i$. Contract the transverse slice of this ball by η about the ambient origin, and increase the s coordinates to s . The resulting transverse ball lies in K_s , proving full dimension. When $m = 0$ this assertion uses the usual convention for $\mathbb{R}^0$.

Finally, for any $a \in \Omega$, the coordinates $s = z - l_a(x)$ and $y = x - a$ make D a 1-model by Lemma 2.1. $\square$

We shall use the following precise consequence of the ellipsoid theorem of John [6]; see also the contact-point formulation and nonsymmetric outer-radius deduction in [1, Section 0]. The normalization is linear and therefore preserves the distinguished origin.

Lemma 2.5. *Let $K \subset \mathbb{R}^d$, $d \geq 1$, be compact and convex with nonempty interior and $0 \in K$. There are an invertible linear map A and a vector c such that*

$$(2.2) \quad \overline{B}_1(c) \subset AK \subset \overline{B}_d(c), \quad |c| \leq d.$$

*In particular $AK \subset \overline{B}_{2d}$; the origin is allowed to lie on ∂K . If in addition $-K \subset C_*K$ for some $C_* \geq 1$, then*

$$(2.3) \quad \overline{B}_{2/(C_*+1)} \subset AK \subset \overline{B}_{2d}.$$

Proof. Choose a maximum-volume inscribed ellipsoid and take the linear part of its normalization to a unit ball. John's Theorem gives the two inclusions in (2.2). Since $0 \in AK$, the outer inclusion implies $|c| \leq d$. The centering assumption also gives $\overline{B}_{1/C_*}(-c/C_*) \subset AK$. The convex combination of this ball and $\overline{B}_1(c)$ with respective weights $C_*/(C_* + 1)$ and $1/(C_* + 1)$ is the centered ball in (2.3). $\square$

By Lemma 2.4, the set of direction numbers achieved by models is a nonempty subset of $\{1, \dots, n + 1\}$. Let k be its maximum.

Proposition 2.6 (Uniform centering at maximal direction number). *Suppose $m = n + 1 - k > 0$. There is a constant $C_* \geq 1$, depending on the family $\mathcal{F}(D)$, such that*

$$(2.4) \quad -K_s \subset C_* K_s \quad (s_i > 0)$$

for every k -model and every admissible choice of its coordinates. In particular, zero is an interior point of every such fiber.

Proof. Suppose that no uniform constant exists. Choose k -models and positive fibers for which the inclusion with constant j fails. Positive diagonal changes in s put those fibers at $\mathbf{1} = (1, \dots, 1)$, and Lemma 2.5 provides linear changes in y . Denote the resulting models by C_j and their fibers at $\mathbf{1}$ by F_j . Thus

$$(2.5) \quad \overline{B}_1(c_j) \subset F_j \subset \overline{B}_{2m}, \quad |c_j| \leq m, \quad -F_j \not\subset jF_j.$$

All these coordinate changes preserve the model properties and the failed inclusion.

For $T \geq 1$ and $0 \leq s_i \leq T$, Lemma 2.4 gives

$$(2.6) \quad K_s^{(j)} \subset K_{T\mathbf{1}}^{(j)} \subset TF_j \subset \overline{B}_{2mT}.$$

Pass to a local limit C and a subsequence with $c_j \rightarrow c$. The limit is contained in the nonnegative s halfspaces, contains the entire orthant at $y = 0$, and has compact caps by (2.6). Moreover,

$$[1, 2]^k \times \overline{B}_1(c_j) \subset C_j,$$

so its limit has nonempty interior. The closure properties of $\mathcal{F}(D)$ show that C is a k -model.

Its fiber $F = K_{\mathbf{1}}$ is the Hausdorff limit of F_j . Only one inclusion needs explanation. If $(s^{(j)}, y_j) \in C_j$ tends to $(\mathbf{1}, y) \in C$, let

$$a_j = \min \left\{ 1, \frac{1}{s_1^{(j)}}, \dots, \frac{1}{s_k^{(j)}} \right\}.$$

The denominators are positive for large j , and $a_j \rightarrow 1$. Contracting by a_j and then increasing the s coordinates gives $(\mathbf{1}, a_j y_j) \in C_j$. Thus every point of F is approximated from the exact fibers F_j ; the other inclusion follows from local convergence. Uniform boundedness makes this Hausdorff convergence. If zero were interior to F , the F_j would eventually contain a fixed ball centered at zero. Their common outer bound would then give a fixed centering constant, contrary to (2.5). Hence $0 \in \partial F$.

A supporting linear functional of F at zero can be made the coordinate y_1 , with $F \subset \{y_1 \geq 0\}$. This inequality holds on all of C : contract any $(s, y) \in C$ until each $s_i \leq 1$, then increase the s coordinates to $\mathbf{1}$ and apply the inequality on F . Since F has nonempty interior, an additional invertible linear choice of transverse coordinates makes an interior point of F equal to $(y_1, y') = (1, 0)$, while retaining the global support $y_1 \geq 0$.

We now construct a model with $k + 1$ directions. Put $d = m - 1$. For $0 < \epsilon < 1/2$, let

$$F_\epsilon = \{y' \in \mathbb{R}^d : (\mathbf{1}, \epsilon, y') \in C\}.$$

This compact slice contains zero in its interior when $d > 0$. Indeed, if $\overline{B}_r((1, 0)) \subset F$, contracting by ϵ about the ambient origin and then adding $(1 - \epsilon)\mathbf{1}$ in the s coordinates shows that

$$\overline{B}_{\epsilon r} \subset F_\epsilon \subset \mathbb{R}^d.$$

When $d > 0$, apply Lemma 2.5 to choose an invertible linear map A_ϵ on $\mathbb{R}^d$ so that

$$\overline{B}_1(c_\epsilon) \subset A_\epsilon F_\epsilon \subset \overline{B}_{2d}, \quad |c_\epsilon| \leq d.$$

If $d = 0$, take A_ϵ to be the unique linear automorphism of $\mathbb{R}^0$ and omit all statements about its balls. Use the affine images

$$C_\epsilon = \{(s, r, z) : (s, \epsilon r, A_\epsilon^{-1} z) \in C\}.$$

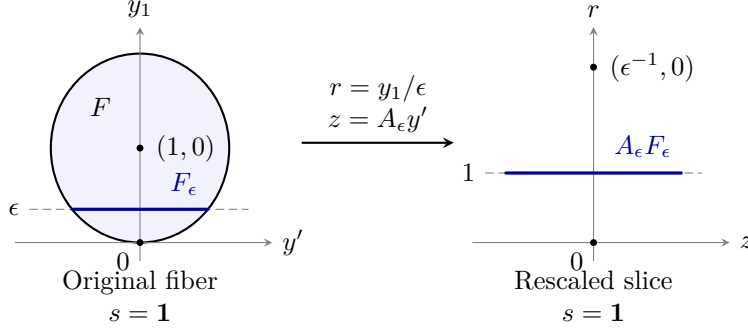


FIGURE 1. The change from (y_1, y') to (r, z) at the fixed base point $s = \mathbf{1}$, shown schematically with one y' coordinate. The actual fiber need not have the smooth or symmetric shape drawn here. The slice $y_1 = \epsilon$ becomes the slice $r = 1$, while A_ϵ normalizes its remaining transverse coordinates. The interior point $(1, 0)$ of F is sent to $(\epsilon^{-1}, 0)$. The right panel shows only this point, the origin, and the distinguished slice; the shape of the transformed fiber is omitted.

They are supported in $s_i, r \geq 0$, contain the origin, and their slice at $(s, r) = (\mathbf{1}, 1)$ has the stated normalization.

Figure 1 illustrates the supporting fiber and the slice used in this rescaling. The remaining step is to turn control of that one slice into a uniform bound for every fixed cap.

Here is a uniform bound for their caps. Suppose $(s, r, z) \in C_\epsilon$ with $s_i, r \leq T$, where $T \geq 1$. Increase s to $T\mathbf{1}$ and contract by $1/T$. This gives

$$(\mathbf{1}, q, z/T) \in C_\epsilon, \quad q = r/T \in [0, 1].$$

The point $(\mathbf{1}, 2, 0)$ also belongs to C_ϵ , since $2\epsilon < 1$ and it is obtained by contracting $(\mathbf{1}, 1, 0) \in C$ and then increasing s . Interpolate these two points with weight

$$a = \frac{1}{2 - q} \in [1/2, 1]$$

on $(\mathbf{1}, q, z/T)$. The resulting point is $(\mathbf{1}, 1, az/T)$, so the normalized slice gives

$$(2.7) \quad |z| \leq 4dT \quad (d > 0).$$

Equivalently, a slice outer radius R gives the bound $2TR$. When $d = 0$, the bounds on s and r themselves suffice. Thus caps in the new nonnegative coordinates are uniformly bounded.

Extract a local limit C_0 as $\epsilon \downarrow 0$, also taking $c_\epsilon \rightarrow c_0$ if $d > 0$. For each fixed $s_i > 0$ and $r \geq 0$, the point $(s, r, 0)$ belongs to C_ϵ as soon as

$$\epsilon r \leq \min\{1, s_1, \dots, s_k\}.$$

For $r > 0$, this follows by contracting $(\mathbf{1}, 1, 0) \in C$ by ϵr and then increasing s ; for $r = 0$ it follows from the original orthant. Hence C_0 contains these points. Closedness adds all faces with some $s_i = 0$, so it contains the entire nonnegative (s, r) orthant at $z = 0$. The support inequalities pass to the limit, and (2.7) makes its caps compact.

For $d > 0$, the limit also contains $\{(\mathbf{1}, 1)\} \times \overline{B}_1(c_0)$. The recession argument in Lemma 2.4, which does not require full dimension, permits addition of every nonnegative (s, r) direction. Thus

$$[1, 2]^{k+1} \times \overline{B}_1(c_0) \subset C_0,$$

which proves full dimension. For $d = 0$, the contained orthant already has full dimension. Finally, affine invariance and diagonal closure of $\mathcal{F}(D)$ give $C_0 \in \mathcal{F}(D)$. It is a $(k+1)$ -model, contradicting maximality.

This contradiction allowed the initial models, fibers, and admissible coordinates to vary arbitrarily. It therefore proves the uniform assertion (2.4). Its last conclusion follows, for example, from (2.3). $\square$

We record the quantitative geometric consequence needed for the later estimates on fixed boxes.

Corollary 2.7. *Under the hypotheses of Proposition 2.6, any chosen positive fiber of any k -model can be moved to $s = \mathbf{1}$ by positive diagonal changes in s and normalized by an invertible linear change in y so that*

$$\overline{B}_{r_0} \subset K_{\mathbf{1}} \subset \overline{B}_{R_0}, \quad r_0 = \frac{2}{C_* + 1}, \quad R_0 = 2m.$$

For every fixed $0 < a < 1 < b$, these normalized models satisfy

$$(2.8) \quad \overline{B}_{ar_0} \subset K_s \subset \overline{B}_{bR_0} \quad (s \in [a, b]^k).$$

Their caps for fixed positive linear forms in s and fixed positive top levels have uniform bounds, and they contain a fixed interior ball about $(\mathbf{1}, 0)$.

For any sequence of affine images of D converging to one of these models, comparable cap and fiber bounds hold eventually, uniformly on each fixed positive box. The constants are independent of the chosen fiber; the required index may depend on its normalization. For finitely many chosen fibers, one index suffices for all of them.

Proof. The normalization follows from Proposition 2.6 and Lemma 2.5. Monotonicity and contraction toward zero give $aK_{\mathbf{1}} \subset K_s \subset bK_{\mathbf{1}}$ on $[a, b]^k$, proving (2.8). If $\ell(s) = \sum_i a_i s_i$ with $a_i > 0$ and $B > 0$, put $T = \max\{1, B/\min_i a_i\}$. Then

$$C \cap \{\ell \leq B\} \subset [0, T]^k \times \overline{B}_{TR_0}.$$

Also $K_s \supset \overline{B}_{r_0/2}$ whenever each $s_i \geq 1/2$, which supplies the fixed interior ball about $(\mathbf{1}, 0)$.

Lemma 2.2 transfers each cap bound to the approximants; the strict point below its top is the origin. For a fixed box $[a, b]^k$, a smaller product $[a, b]^k \times \overline{B}_{ar_0/2}$ is a compact subset of the interior of the model, as is seen by first enlarging the positive s box slightly. Interior convergence therefore gives the lower fiber bound on the entire box, and a containing cap gives the upper bound. These constants depend only on the displayed fixed parameters. Taking the maximum of the finitely many necessary indices proves the final statement. $\square$

3. AFFINE AREA AND A CAP IDENTITY

From this section onward, the original graph also satisfies Equation (1.1). We derive stationarity in arbitrary affine coordinates and a cap identity that will control the inverse-Hessian weight in the logarithmic cell estimate. The affine-area variational framework is due to Calabi [4]; the formulas needed here are derived explicitly. Throughout,

$$\delta = \frac{1}{n+2}.$$

For a smooth locally strictly convex hypersurface parametrized by X , an *inward conormal* is a covector ν annihilating its tangent space and pointing into the convex body. Its second form is $H_{ij} = \nu X_{ij}$; we choose the sign so that H is positive definite. If $\nu\xi = 1$, its affine area density is

$$(3.1) \quad dA = (\det H)^\delta |\det(X_1, \dots, X_n, \xi)|^{n\delta} dx_1 \cdots dx_n.$$

Changing ξ by a tangent vector leaves the determinant unchanged. Multiplying ν by a positive scalar multiplies the two factors by reciprocal powers. A parameter change contributes the Jacobian to the power $2\delta + n\delta = 1$, so (3.1) is a well-defined density. For the graph $X(x) = (x, u(x))$, choose $\nu = (-Du, 1)$, $\xi = (0, 1)$; then

$$dA = (\det D^2 u)^\delta dx.$$

An invertible affine map with linear part L replaces the conormal by νL^{-1} . The second form is unchanged and the tangential determinant acquires a factor $\det L$. Thus

$$(3.2) \quad dA_{L(M)} = |\det L|^{n\delta} dA_M.$$

With the unit inward normal, the same formula reads $dA = K^\delta dS$, where K is Gauss curvature and dS is Euclidean hypersurface area.

Lemma 3.1 (Stationarity). *The equation in Theorem 1.1 implies stationarity of affine area under every smooth compactly supported variation of the graph. Every invertible affine image of the graph is stationary under such variations as well.*

Proof. The Hessian cofactor matrix satisfies $\partial_i U^{ij} = \partial_j U^{ij} = 0$. This follows by differentiating its alternating determinant formula: terms cancel in pairs by symmetry of the third derivatives. Since

$$(\det D^2 u)^\delta (D^2 u)^{-1} = wU,$$

the first variation in a compactly supported graph direction η is

$$\delta \int_{\Omega} \operatorname{tr}((D^2 u)^{-1} D^2 \eta) dA = \delta \int_{\Omega} \eta U^{ij} w_{ij} dx = 0.$$

Here and below repeated coordinate indices are summed. For a compactly supported parametrized variation, its normal part can be represented locally as a graph variation, while its tangential part is a reparametrization. A partition of unity therefore gives stationarity for parametrized variations. Formula (3.2) transfers this statement to every affine image. $\square$

Lemma 3.2 (Area in a ball). *For each n and $R > 0$, the affine area of the part in B_R of any smooth positively curved convex boundary in $\mathbb{R}^{n+1}$ is bounded by a constant depending only on n, R .*

Proof. The Gauss map is injective: two distinct contact points with the same supporting normal would force a boundary segment in their supporting hyperplane, contrary to positive curvature. Thus $\int K dS$ over the part in the ball is at most the area of S^n .

For the Euclidean area bound, cover this part by the $2(n+1)$ sets on which some signed component of the unit normal is at least $(n+1)^{-1/2}$. On each set, convexity makes projection to the corresponding coordinate hyperplane injective: points with a given projection lie on a line meeting the body in an interval, and the chosen sign selects one endpoint. The projection Jacobian is at least $(n+1)^{-1/2}$, and its image lies in the radius- R ball. This bounds $\int dS$. Finally, Hölder's inequality gives

$$\int K^\delta dS \leq \left(\int K dS \right)^\delta \left(\int dS \right)^{1-\delta}.$$

$\square$

Lemma 3.3 (Cap identity). *Let M be an affine image of the original graph, bounding its convex body D' . Let $o \in \operatorname{int} D'$, let ℓ be an affine function on the ambient space, and let $\psi \in C^\infty(\mathbb{R})$. Suppose that the restrictions of $\psi(\ell), \psi'(\ell), \psi''(\ell)$ vanish outside a compact subset of M . Then*

$$(3.3) \quad \int_M \left[\nu(o - X) |d\ell|_{H^{-1}}^2 \psi''(\ell) + n(\ell(o) - \ell(X)) \psi'(\ell) - n(n+1) \psi(\ell) \right] dA = 0.$$

Here $d\ell$ is restricted to the tangent space. The first term does not depend on the scaling of the inward conormal.

Proof. The identity is affine invariant up to the common area factor in (3.2), so it suffices to work on the original graph. Put $H = D^2 u$, write $o = (o', o_{n+1})$, and use $\nu = (-Du, 1)$. Set

$$q(x) = \ell(x, u(x)), \quad Z(x) = \nu(o - X) = o_{n+1} - u + (x - o') \cdot Du.$$

If ℓ_z denotes the ambient vertical coefficient of ℓ , then

$$DZ = H(x - o'), \quad \text{tr}(H^{-1}D^2Z) = n + (x - o') \cdot D \log \det H, \quad D^2q = \ell_z H.$$

Testing Lemma 3.1 with $Z\psi(q)$ gives

$$0 = \int_M [Z|dq|_{H^{-1}}^2 \psi''(q) + nZ\ell_z \psi'(q) + 2(x - o') \cdot Dq \psi'(q) + n\psi(q) + \psi(q)(x - o') \cdot D \log \det H] dA.$$

Because $dA = (\det H)^\delta dx$, integration by parts in the last term replaces it by

$$-\delta^{-1} [n\psi(q) + (x - o') \cdot Dq \psi'(q)].$$

The coefficient of ψ is consequently $-n(n+1)$. The coefficient of ψ' is

$$n[Z\ell_z - (x - o') \cdot Dq] = n[\ell(o) - q].$$

This is (3.3). Compact support justifies both integrations by parts. $\square$

Remark 3.4. For later use, ψ may be convex and zero above a fixed cap height, without having compact support as a function on $\mathbb{R}$. If that cap is compact, its restriction to M satisfies the support condition in Lemma 3.3. Also, if $B_r(o) \subset D'$, the supporting-plane inequality gives

$$(3.4) \quad \nu(o - X) \geq r|\nu| \quad (X \in M).$$

4. AN INTEGRAL INEQUALITY FOR CONVEX TUBES

Let $E \subset \mathbb{R}^{n+1}$ be a closed full-dimensional convex set whose boundary is smooth, has positive second fundamental form, and is stationary for affine area. In the applications, E will be an affine image of the original epigraph. Fix a splitting

$$\mathbb{R}^{n+1} = \mathbb{R}_s^k \times \mathbb{R}_y^m, \quad 2 \leq k \leq n, \quad m = n + 1 - k,$$

and an open box $\mathcal{B} \subset (0, \infty)^k$. Suppose that for every $s \in \mathcal{B}$ the fiber

$$K_s = \{y \in \mathbb{R}^m : (s, y) \in E\}$$

is compact and contains 0 in its interior. Our goal is the logarithmic inequality in Proposition 4.1, whose constant depends only on n , independently of the box and the shapes of its fibers.

4.1. Support coordinates and invariant measures. Fibers over compact subsets of $\mathcal{B}$ are uniformly bounded. Otherwise there would be $(s_j, y_j) \in E$ with s_j bounded and $|y_j| \rightarrow \infty$. After passage to a subsequence, $y_j/|y_j| \rightarrow e \in S^{m-1}$. Fix $s_0 \in \mathcal{B}$, so that $(s_0, 0) \in E$. For every fixed $t \geq 0$, convexity and closedness give

$$(s_0, te) = \lim_{j \rightarrow \infty} \left[\left(1 - \frac{t}{|y_j|}\right) (s_0, 0) + \frac{t}{|y_j|} (s_j, y_j) \right] \in E.$$

This contradicts compactness of K_{s_0} . Thus the part of E over any compact base subset is compact.

The projection of ∂E to the s variables has full rank above $\mathcal{B}$. Indeed, otherwise a conormal at such a point would have zero y component. Its supporting hyperplane would then give a supporting hyperplane to the projection of E at an interior point of that projection, which is impossible. Thus ∂K_s is smooth, and the restriction of the second fundamental form makes it strictly positively curved. Its outward Gauss map is a diffeomorphism onto S^{m-1} when $m > 1$. The Gauss maps also depend smoothly on s , since their angular differentials are invertible. Consequently the support function $h(s, e) = \max_{y \in K_s} e \cdot y$ is smooth and positive on

$$\mathcal{T} = \mathcal{B} \times S^{m-1},$$

and the corresponding part of ∂E is parametrized by

$$(4.1) \quad X(s, e) = (s, Y(s, e)), \quad Y = he + \nabla_S h.$$

Here ∇_S is the round connection. This support-function reconstruction and its angular matrix are classical; see, for example, [11, preprint version, Section 7, Equations (7.15)–(7.17)]. We compute below how they enter the second form and affine-area density of the present tube. When $m = 1$, we regard $S^0 = \{-1, 1\}$ as the two outward directions of the interval K_s ; Equation (4.1) then means $Y = he$ on the two endpoint sheets. Throughout this section, angular matrices in this case have size zero, their determinants equal 1, and angular derivatives, traces, and contractions equal 0. The measure $d\omega$ on S^0 is counting measure.

Set

$$B = -D_s^2 h, \quad R = \nabla_S^2 h + h \text{Id}, \quad \nu = (D_s h, -e).$$

The identities $e \cdot Y = h$, $e \cdot Y_s = h_s$, and $e \cdot d_e Y = 0$ show that ν annihilates the tangent vectors of X . It is inward because the fiber lies on the side $e \cdot y \leq h(s, e)$. Differentiating the same identities gives

$$\nu X_{s_i s_j} = -h_{s_i s_j}, \quad \nu X_{s_i e_a} = 0, \quad \nu X_{e_a e_b} = R_{ab}.$$

For the last equality we used $d_e Y = R$ in round orthonormal frames. Thus the second form for this conormal is

$$(4.2) \quad H = \text{diag}(B, R).$$

Both blocks are positive definite. Taking $\xi = (0, -e)$ gives $\nu \xi = 1$ and $|\det(X_1, \dots, X_n, \xi)| = \det R$. The affine-area formula therefore becomes

$$(4.3) \quad dA = I_0 \, ds \, d\omega, \quad I_0 = (\det B)^\delta (\det R)^{1-\delta},$$

where we used $(n+1)\delta = 1 - \delta$.

To compare fibers after linear normalizations, we also need a measure invariant under separate linear changes of s and y . The vector $V = (0, -Y)$ points from the fiber boundary toward its origin, and $\nu V = h > 0$. Normalizing the conormal to take value 1 on V therefore defines the positive metric

$$G = H/h.$$

Its Riemannian volume is

$$(4.4) \quad d\mu = \mu_0 \, ds \, d\omega, \quad \mu_0 = h^{-n/2} \sqrt{\det B \det R}.$$

This definition makes G and μ invariant under separate invertible linear changes of s and y . To check the normalization explicitly, let $(s', y') = (As, Ly)$ and, for an old unit normal e , put

$$t = |L^{-\top} e|, \quad e' = L^{-\top} e/t.$$

The new support function obeys $h'(As, e') = h(s, e)/t$, and its conormal is $\text{diag}(A, L)^{-\top} \nu/t$. Under the induced parameter map the new second form pulls back to H/t . Dividing it by $h' = h/t$ recovers G , including the change in angular parameters.

For s covectors we henceforth use the inner product and norm given by hB^{-1} , and for angular covectors those given by hR^{-1} . We write both as $\langle \cdot, \cdot \rangle$ and $|\cdot|$, with the variables specifying the block. Thus, for example,

$$|D_s g|^2 = h (D_s g)^\top B^{-1} D_s g.$$

All unqualified integrals below are over $\mathcal{T}$. Every cutoff in the s variables has compact support in $\mathcal{B}$, so these integrals and the corresponding variations have compact support on the hypersurface.

4.2. The stationarity equation in support coordinates. The support coordinates have identified both affine area and the invariant volume. We next express stationarity through their density ratio. This supplies the differential identity whose weighted integral will have a strictly positive quadratic part in dimensions three through nine.

We first record the cofactor identities needed for integration by parts. Since $B = D_s^2(-h)$, its cofactor is divergence free in the flat variables. The angular tensor R is Codazzi:

$$\nabla_a R_{bc} = \nabla_b R_{ac}.$$

One way to see this is to differentiate $d_e Y = R$ in the ambient Euclidean space. Commuting the two derivatives of Y and taking tangential components gives the displayed identity; the normal components cancel because R is symmetric. Equivalently, the curvature terms from commuting third derivatives of h cancel the derivatives of $h\text{Id}$.

For completeness, if A is any symmetric Codazzi tensor of positive size r , its cofactor is divergence free. In normal orthonormal frames the cofactor is

$$(\text{cof } A)^{ij} = \frac{1}{(r-1)!} \delta_{jj_2 \dots j_r}^{ii_2 \dots i_r} A_{i_2 j_2} \cdots A_{i_r j_r},$$

where the symbol on the right is the alternating Kronecker symbol. On differentiating and contracting the derivative index with i , each resulting term contains $\nabla_i A_{i_a j_a} = \nabla_{i_a} A_{i j_a}$, symmetric in two indices against which that symbol is alternating. Each term is therefore zero. Symmetry gives the divergence identity in either contracted index. For $r = 1$ the cofactor is the constant 1; for the empty angular block no identity is needed. In particular,

$$(4.5) \quad \partial_{s_i}(\text{cof } B)^{ij} = 0, \quad \nabla_a(\text{cof } R)^{ab} = 0.$$

Define

$$(4.6) \quad a = \left(\frac{\det B}{\det R} \right)^\delta, \quad f = \log(a/h), \quad c = \frac{n+2}{2}.$$

The function f records the ratio of affine area to invariant volume: by Equations (4.3) and (4.4),

$$\frac{dA}{d\mu} = h^{n/2} \left(\frac{\det B}{\det R} \right)^{-n/(2(n+2))} = \left(\frac{a}{h} \right)^{-n/2} = e^{-nf/2}.$$

A variation $h + \varepsilon \eta$, with $\eta \in C_c^\infty(\mathcal{T})$, gives a compactly supported smooth variation of X . Positivity persists for sufficiently small ε on its compact support. Differentiating Equation (4.3), using stationarity, and dividing by δ gives

$$0 = \int \left[-I_0 B^{-1} : D_s^2 \eta + (n+1) I_0 R^{-1} : (\nabla_S^2 \eta + \eta \text{Id}) \right] ds \, d\omega.$$

The colon denotes contraction in the indicated orthonormal frames. The scalar factors in this expression satisfy

$$I_0 B^{-1} = (\text{cof } B) a^{-(n+1)}, \quad I_0 R^{-1} = (\text{cof } R) a.$$

Two integrations by parts using Equation (4.5) therefore yield

$$(4.7) \quad -(\text{cof } B) : D_s^2(a^{-(n+1)}) + (n+1)(\text{cof } R) : (\nabla_S^2 a + a \text{Id}) = 0.$$

There is no angular boundary term. In the case $m = 1$, the calculation is performed separately on the two endpoint sheets and the second term is absent.

Writing $t = \log a$, we have

$$\begin{aligned} D_s^2(a^{-(n+1)}) &= (n+1)a^{-(n+1)}(-D_s^2 t + (n+1)D_{st} \otimes D_{st}), \\ \nabla_S^2 a &= a(\nabla_S^2 t + \nabla_{St} \otimes \nabla_{St}). \end{aligned}$$

Dividing Equation (4.7) by $(n+1)I_0$ gives

$$(4.8) \quad \begin{aligned} & B^{-1} : D_s^2 \log a - (n+1)(D_s \log a)^\top B^{-1} D_s \log a \\ & + R^{-1} : \nabla_S^2 \log a + (\nabla_S \log a)^\top R^{-1} \nabla_S \log a + \text{tr } R^{-1} = 0. \end{aligned}$$

Introduce

$$\begin{aligned} p &= D_s \log h, & d &= D_s f, & p_e &= \nabla_S \log h, & d_e &= \nabla_S f, \\ T_s g &= h B^{-1} : D_s^2 g, & T_e g &= h R^{-1} : \nabla_S^2 g. \end{aligned}$$

Direct differentiation of h gives

$$(4.9) \quad T_s \log h = -k - |p|^2, \quad T_e \log h = n - k - h \text{tr } R^{-1} - |p_e|^2.$$

Substitute $\log a = \log h + f$ in Equation (4.8) and multiply by h . The angular trace and $|p_e|^2$ terms cancel, and Equation (4.9) gives

$$(4.10) \quad n - 2k + T_s f - |p|^2 - (n+1)|p + d|^2 + T_e f + 2\langle p_e, d_e \rangle + |d_e|^2 = 0.$$

4.3. Weighted integration and coercivity. Equation (4.10) contains derivatives with different drifts in the base and angular variables. We integrate with the invariant measure μ , retaining both drifts until the cancellation in Equation (4.15). The dimension restriction enters only in the quadratic form that remains.

The matrix densities for T_s and T_e relative to $ds \, d\omega$ are

$$(4.11) \quad \mu_0 h B^{-1} = (\text{cof } B) h^{-n} e^{-cf}, \quad \mu_0 h R^{-1} = (\text{cof } R) h^2 e^{cf}.$$

For example, the first follows from $\sqrt{\det R / \det B} = a^{-c}$ and $h^{1-n/2} a^{-c} = h^{-n} e^{-cf}$; the second follows in the same way with the signs of the exponents reversed.

Let $v \in C_c^\infty(\mathcal{B})$. The logarithmic derivatives of the scalar factors in Equation (4.11) are $-np - cd$ and $2p_e + cd_e$, respectively. One integration by parts and Equation (4.5) therefore give, for every smooth g on $\mathcal{T}$,

$$(4.12) \quad \begin{aligned} \int v^2 T_s g \, d\mu &= \int \left[v^2 \langle np + cd, D_s g \rangle - 2v \langle D_s v, D_s g \rangle \right] d\mu, \\ \int v^2 T_e g \, d\mu &= - \int v^2 \langle 2p_e + cd_e, \nabla_S g \rangle d\mu. \end{aligned}$$

The angular identity has no cutoff term because v depends only on s . Taking $g = \log h$ in the first identity and using Equation (4.9) yields

$$(4.13) \quad \int v^2 \left[k + (n+1)|p|^2 + c \langle d, p \rangle \right] d\mu = 2 \int v \langle D_s v, p \rangle d\mu.$$

Next integrate Equation (4.10) against v^2 and apply Equation (4.12) with $g = f$. Expanding the squares gives

$$(4.14) \quad \begin{aligned} \int v^2 \left[\frac{n}{2}(|d|^2 + |d_e|^2) + (n+2)(|p|^2 + \langle p, d \rangle) - (n-2k) \right] d\mu \\ = -2 \int v \langle D_s v, d \rangle d\mu. \end{aligned}$$

In particular, subtracting twice Equation (4.13) gives the exact cancellation

$$(4.15) \quad \begin{aligned} \frac{n}{2} \int v^2 (|d|^2 + |d_e|^2 - 2|p|^2 - 2) d\mu &= -2 \int v \langle D_s v, d \rangle d\mu \\ &\quad - 4 \int v \langle D_s v, p \rangle d\mu. \end{aligned}$$

We combine these identities to control both the μ -mass and the derivative terms. For a parameter $\lambda > 0$, add $2\lambda/n$ times Equation (4.15) to Equation (4.13). The resulting left integrand is

$$F_\lambda = k - 2\lambda + (n + 1 - 2\lambda)|p|^2 + c\langle p, d \rangle + \lambda(|d|^2 + |d_e|^2),$$

and the identity is

$$(4.16) \quad \int v^2 F_\lambda \, d\mu = \int v \left\langle D_s v, \left(2 - \frac{8\lambda}{n}\right) p - \frac{4\lambda}{n} d \right\rangle d\mu.$$

In an orthonormal frame for the s covectors, the coefficient matrix for each paired component of p, d is

$$\begin{pmatrix} n + 1 - 2\lambda & (n + 2)/4 \\ (n + 2)/4 & \lambda \end{pmatrix}.$$

At $\lambda = 1$, its determinant is

$$n - 1 - \frac{(n + 2)^2}{16} = \frac{(n - 2)(10 - n)}{16}.$$

This is positive for $3 \leq n \leq 9$, but the scalar term $k - 2$ vanishes when $k = 2$. Choosing λ slightly below one makes the scalar term positive while preserving the positive quadratic form. For the rest of the argument impose $3 \leq n \leq 9$ and fix $\lambda = 15/16$, which works throughout this range. Then $k - 2\lambda \geq 1/8$, and the determinant satisfies

$$\lambda(n + 1 - 2\lambda) - \frac{(n + 2)^2}{16} = \frac{-8n^2 + 88n - 137}{128} \geq \frac{7}{128} \quad (3 \leq n \leq 9),$$

where the last inequality follows by concavity of the quadratic and its endpoint values. Both diagonal entries are positive. Thus there is a constant $\kappa_n > 0$ such that

$$F_\lambda \geq \kappa_n(1 + |p|^2 + |d|^2 + |d_e|^2).$$

Young's inequality absorbs the right side of Equation (4.16), giving

$$(4.17) \quad \int v^2(1 + |p|^2 + |d|^2 + |d_e|^2) \, d\mu \leq C_n \int |D_s v|^2 \, d\mu.$$

All constants here depend only on n ; the estimates are uniform for $2 \leq k \leq n$.

4.4. The logarithmic inequality. For $1 \leq i \leq k$, put $\alpha_i = \log s_i$ and define the measure

$$(4.18) \quad d\sigma = \left(1 + \sum_{i=1}^k |D_s \log s_i|^2\right) d\mu, \quad |D_s \log s_i|^2 = \frac{h(B^{-1})_{ii}}{s_i^2}.$$

This measure is invariant under invertible linear changes in y and positive diagonal changes in s : the metric G and its volume are invariant, while each α_i changes only by an additive constant.

Proposition 4.1. *Let $3 \leq n \leq 9$ and $2 \leq k \leq n$. On any tube satisfying the hypotheses of this section, there is a constant C_n , depending only on n , such that every $\tilde{v} \in C_c^\infty(\log \mathcal{B})$ satisfies*

$$(4.19) \quad \int_{\mathcal{T}} \tilde{v}(\log s)^2 \, d\sigma \leq C_n \int_{\mathcal{T}} |\nabla \tilde{v}(\log s)|_{\mathbb{R}^k}^2 \, d\sigma.$$

Here $\log s = (\log s_1, \dots, \log s_k)$ and the gradient on the right is Euclidean. No completeness of the metric G is required.

Proof. Since $T_s \alpha_i = -|D_s \alpha_i|^2$, Equation (4.12) gives

$$\begin{aligned} \int v^2 |D_s \alpha_i|^2 \, d\mu &= - \int v^2 \langle np + cd, D_s \alpha_i \rangle \, d\mu \\ &\quad + 2 \int v \langle D_s v, D_s \alpha_i \rangle \, d\mu. \end{aligned}$$

Young's inequality, followed by Equation (4.17), therefore gives

$$\int v^2 |D_s \alpha_i|^2 d\mu \leq C_n \int |D_s v|^2 d\mu.$$

Summing these inequalities and the term controlling $\int v^2 d\mu$ in Equation (4.17), we obtain

$$\int v^2 d\sigma \leq C_n \int |D_s v|^2 d\mu.$$

Finally take $v(s) = \tilde{v}(\log s)$. The chain rule and Cauchy–Schwarz in the positive inner product hB^{-1} imply

$$\begin{aligned} |D_s v|^2 &= \left| \sum_{i=1}^k (\partial_i \tilde{v})(\log s) D_s \alpha_i \right|^2 \\ &\leq |\nabla \tilde{v}(\log s)|_{\mathbb{R}^k}^2 \sum_{i=1}^k |D_s \alpha_i|^2. \end{aligned}$$

Equation (4.19) follows from Equation (4.18). Every integration used a compactly supported cutoff on the smooth tube, which also proves the last assertion. $\square$

5. EXCLUDING MODELS WITH SEVERAL DIRECTIONS

We now combine the logarithmic inequality with estimates on smooth approximants of a model. All curvature quantities in this section belong to the approximants; no differentiability of the limiting body is used.

Proposition 5.1. *Suppose $3 \leq n \leq 9$. The largest number of directions of a model in $\mathcal{F}(D)$, in the sense of Definition 2.3, is one.*

We first establish the area estimate that prevents the measures in the logarithmic inequality from vanishing. The comparison uses concavity of the affine-area integrand and stationarity, as in the local maximization argument of Trudinger and Wang [9, Section 6, Equations (6.2)–(6.4)]. Here the comparison is applied to uniformly normalized caps of smooth approximants.

Lemma 5.2. *Let D_j be affine images of D converging locally to a full-dimensional closed convex set C . Suppose that a nonzero linear function ℓ and a number b satisfy*

$$C \cap \{\ell \leq b\} \text{ is compact,} \quad \text{int } C \cap \{\ell < b\} \neq \emptyset.$$

There are $c > 0$ and j_0 such that

$$\int_{\partial D_j \cap \{\ell < b\}} dA_j \geq c \quad (j \geq j_0).$$

Proof. Lemma 2.2 gives a uniform bound for these caps. Choose a closed ball $\overline{B}_r(o)$ in $\text{int } C \cap \{\ell < b\}$. After decreasing r , the same ball lies in D_j for all sufficiently large j . Fix $\varepsilon_0 > 0$ with $\ell(o) \leq b - \varepsilon_0$.

Let e_j be the unit vector in the image of the original upward vertical direction under the affine map defining D_j . It is a recession direction of D_j , and ∂D_j is graphical along it. The component $\ell(e_j)$ is bounded below by a positive constant. Indeed, the ray $o + te_j$ lies in D_j ; if $\ell(e_j) \leq 0$, the whole ray lies in the cap, and otherwise its initial segment of length $(b - \ell(o))/\ell(e_j)$ lies there. Uniform boundedness of the caps excludes both a nonpositive component and components tending to zero. Consequently

$$V_j = \frac{e_j}{\ell(e_j)} \quad \text{satisfies} \quad \ell(V_j) = 1, \quad |V_j| \leq C.$$

Use the coordinates

$$X = \zeta + \tau V_j, \quad \zeta \in \ker \ell.$$

These coordinate maps and their inverses are uniformly bounded. With an orthonormal basis of $\ker \ell$, their determinants have absolute value $1/|\ell|$, where $|\ell|$ is the Euclidean norm of the covector. In these coordinates ∂D_j is the graph of a smooth strictly convex function g_j , and the cap corresponds to $\tau \leq b$.

The sets

$$E_j = \{\zeta : g_j(\zeta) < b\}$$

are uniformly bounded. Every closed sublevel of g_j at a level at most b is compact within its domain: its lifted graph lies in a compact cap, and the closed hypersurface ∂D_j is precisely the graph in this direction. Writing $\zeta_j = o - \ell(o)V_j$, the common interior ball gives

$$g_j(\zeta) \leq b - \varepsilon_0 \quad (\zeta \in B_{\ker \ell}(\zeta_j, r)).$$

Here $B_{\ker \ell}(\zeta_j, r)$ is the radius- r ball in $\ker \ell$ centered at ζ_j . The centers ζ_j and the sets E_j are uniformly bounded. We may therefore choose one $\alpha > 0$ so small that the quadratics

$$q_j(\zeta) = b - \frac{2\varepsilon_0}{3} + \alpha|\zeta - \zeta_j|^2$$

satisfy

$$b - \frac{2\varepsilon_0}{3} \leq q_j \leq b - \frac{\varepsilon_0}{3} \quad \text{on } E_j.$$

Let χ be a smooth convex regularization of the positive part, with $0 \leq \chi' \leq 1$, equal to 0 for arguments at most $-\eta$ and equal to the argument for arguments at least η , where $0 < \eta < \varepsilon_0/6$. On E_j set

$$\tilde{g}_j = g_j + \chi(q_j - g_j).$$

This equals q_j on the indicated base ball and equals g_j near the level b . Extending by g_j outside E_j gives a smooth perturbation whose difference from g_j has compact support in E_j . Its Hessian is positive definite, since it is a convex combination of $D^2 g_j$ and $D^2 q_j$, plus a positive semidefinite matrix.

The function $F(H) = (\det H)^\delta$ is concave on positive definite matrices: the determinant n -th root is concave, and $n\delta = n/(n+2) < 1$. Concavity and Lemma 3.1 give

$$\begin{aligned} \int_{E_j} F(D^2 \tilde{g}_j) \, d\zeta &\leq \int_{E_j} F(D^2 g_j) \, d\zeta \\ &\quad + \delta \int_{E_j} F(D^2 g_j) (D^2 g_j)^{-1} : D^2(\tilde{g}_j - g_j) \, d\zeta \\ &= \int_{E_j} F(D^2 g_j) \, d\zeta. \end{aligned}$$

The first integral is bounded below by $|B_{\ker \ell}(0, r)|(2\alpha)^{n\delta} > 0$. Affine covariance of area, with the uniformly controlled coordinate determinants above, proves the assertion. $\square$

5.1. Positive mass away from the coordinate faces. Fix a model C with the maximal number k of directions, and choose affine images $D_j \rightarrow C$. Write its coordinates as $(s, y) \in \mathbb{R}^k \times \mathbb{R}^m$, where $m = n + 1 - k$. Choose $b > 0$ so that the cap for $\ell_0(s) = s_1 + \cdots + s_k$ contains an interior ball strictly below its top. Such a choice is possible because C has nonempty interior. The cap is compact by Definition 2.3. Lemma 5.2 gives

$$(5.1) \quad A_j(\partial D_j \cap \{\ell_0 < b\}) \geq c > 0$$

for large j , and all these caps lie in a fixed ball.

For any fixed $0 < \varepsilon < 1$, convergence and the uniform cap bound imply that $s_i \geq -\varepsilon$ everywhere on the approximant caps once j is sufficiently large. In the part where $s_i \leq \varepsilon$, stretch the single coordinate s_i by $1/\varepsilon$. The transformed boundary part lies in a fixed ball, independent of ε and j . Lemma 3.2 and the $|\det L|^{n\delta}$ covariance of affine area give

$$(5.2) \quad A_j(\partial D_j \cap \{\ell_0 < b, s_i \leq \varepsilon\}) \leq C\varepsilon^{n\delta}.$$

Only the boundary part under consideration needs to lie in the ball. Choose ε so small that the sum of these k bounds is less than $c/2$, and then choose j large enough. Equations (5.1) and (5.2) leave area at least $c/2$ in a fixed bounded region where all $s_i > \varepsilon$.

If $m = 0$, the model is the nonnegative orthant in $\mathbb{R}^{n+1}$. The remaining region is contained in a compact subset of its interior, which lies in $\text{int } D_j$ for large j . It cannot contain any points of ∂D_j , a contradiction. Hence

$$(5.3) \quad k \leq n, \quad m \geq 1.$$

For the rest of the section, suppose that $k \geq 2$. On every fixed compact box of positive s 's, the fibers of D_j are uniformly bounded and contain zero in their interiors for large j . This follows from Proposition 2.6, Lemma 2.4, and Lemma 2.2. Thus the tube coordinates and measures (4.3), (4.4), and (4.18) are available there. For a base set $E \subset (0, \infty)^k$, write $\mu_j(E)$ and $\sigma_j(E)$ for the masses of $E \times S^{m-1}$. In particular, define

$$Q_L = \{s \in (0, \infty)^k : |\log s|_\infty \leq L\}, \quad \log s = (\log s_1, \dots, \log s_k).$$

The preceding area bound gives a fixed positive box Q such that $\int_{Q \times S^{m-1}} I_0 \, ds \, d\omega \geq c/2$ for large j . On this box h is uniformly bounded above, and

$$\int_{Q \times S^{m-1}} \det R \, ds \, d\omega \leq C.$$

Indeed, $\det R \, d\omega$ is Euclidean area on the boundary of a uniformly bounded convex fiber. When $m = 1$, it is counting measure on the two endpoints, with $\det R = 1$. Using the density μ_0 from (4.4), the exact factorization

$$I_0 = \mu_0^{2\delta} h^{n\delta} (\det R)^{1-2\delta}$$

and Hölder's inequality yield

$$\frac{c}{2} \leq \int_{Q \times S^{m-1}} I_0 \, ds \, d\omega \leq C\mu_j(Q)^{2\delta}.$$

Since $\sigma_j \geq \mu_j$, enlarging Q to a fixed Q_{L_0} proves that there are $L_0, c_0 > 0$ with

$$(5.4) \quad \sigma_j(Q_{L_0}) \geq c_0 \quad \text{for all sufficiently large } j.$$

5.2. A uniform bound on each logarithmic cell. We now have positive mass in one fixed logarithmic box. To control growth, we need a bound for the mass of every cell that is independent of its logarithmic position. Centering provides this uniformity after affine normalization. The cap identity controls an inverse-Hessian integral, and bounds for Hessian minors provide the complementary factor in a weighted interpolation.

Lemma 5.3. *There is a constant C_{cell} , independent of $a \in \mathbb{R}^k$, such that*

$$\sigma_j \left\{ s : \frac{1}{2}e^{a_i} \leq s_i \leq 2e^{a_i} \text{ for every } i \right\} \leq C_{\text{cell}}$$

for every sufficiently large j . The threshold for j may depend on a .

Proof. Rescale s_i by e^{-a_i} and make an invertible linear change in y . By Corollary 2.7, these changes can be chosen so that the limit fiber above $\mathbf{1} = (1, \dots, 1)$ satisfies

$$B_{r_*} \subset K_1 \subset B_{R_*},$$

where $r_*, R_* > 0$ are independent of a . We use the same affine change on the approximants. The measures σ_j are preserved by these normalizations.

Corollary 2.7 also gives uniform fiber bounds on the fixed box $[1/4, 4]^k$, uniform bounds for caps with fixed positive linear forms and fixed top levels, and a fixed interior ball centered at $o = (\mathbf{1}, 0)$. Applying its approximation statement on a slightly larger positive box gives

$$(5.5) \quad 0 < c_1 \leq h \leq C_1 \quad \text{on } [1/4, 4]^k \times S^{m-1}$$

and the same uniform cap and interior-ball bounds for all sufficiently large j . These constants are independent of a ; only the required index depends on the normalization.

Let $Q = [1/2, 2]^k$ and $\mathcal{T} = Q \times S^{m-1}$. By (4.18) and (5.5),

$$\sigma_j(Q) \leq C \int_{\mathcal{T}} \sqrt{\det B \det R} (1 + \operatorname{tr} B^{-1}) \, ds \, d\omega,$$

since $s_i^{-2} \leq 4$ on Q . To control the inverse-Hessian factor in this integral, we first prove

$$(5.6) \quad \int_{\mathcal{T}} I_0 \operatorname{tr} B^{-1} \, ds \, d\omega \leq C.$$

For $r = 1, \dots, k$, set

$$a^{(r)} = \mathbf{1} + e_r, \quad \ell_r(s) = a^{(r)} \cdot s.$$

Their values on Q lie in $J = [(k+1)/2, 2(k+1)]$. Choose a smooth nonnegative function ρ , compactly supported below a fixed number $b_0 > 2(k+1)$, with $\rho \geq 1$ on J , and put

$$\psi(t) = \int_t^{b_0} (r-t)\rho(r) \, dr.$$

Then $\psi'' = \rho \geq 0$, and ψ vanishes above b_0 . Although it is affine sufficiently far below, its surface test is compactly supported in the bounded cap $\ell_r \leq b_0$. The functions ψ, ψ' and the levels ℓ_r are bounded there by uniform constants.

Apply Lemma 3.3 with the center o of the fixed interior ball. If its radius is r_0 , the supporting-plane inequality gives

$$\nu(o - X) \geq r_0 |\nu|.$$

The integrand involving ψ'' in that identity is nonnegative over the whole cap. On the tube, take the conormal $\nu = (D_s h, -e)$; it has norm at least one, and

$$|d\ell_r|_{H^{-1}}^2 = (a^{(r)})^\top B^{-1} a^{(r)}.$$

The other terms in the cap identity have bounded integrals by Lemma 3.2 and the uniform cap bound. Hence

$$\int_{\mathcal{T}} I_0 (a^{(r)})^\top B^{-1} a^{(r)} \, ds \, d\omega \leq C.$$

Summing over r proves (5.6), since

$$\sum_{r=1}^k a^{(r)} (a^{(r)})^\top = \operatorname{Id} + (k+2) \mathbf{1} \mathbf{1}^\top \geq \operatorname{Id}.$$

We next claim that

$$(5.7) \quad \int_{\mathcal{T}} \det B (1 + \operatorname{tr} B^{-1}) \, ds \, d\omega \leq C.$$

For each fixed e , the function $-h(\cdot, e)$ is strictly convex. Its gradient is uniformly bounded on Q by (5.5) on the larger box. For a principal minor, hold the unused s -coordinates fixed and consider the gradient in the selected coordinates. It is injective, because the corresponding Hessian is positive definite, and its image lies in a fixed bounded box. The change-of-variables formula bounds the integral of its Jacobian, which is precisely that principal minor. Integrating

in the unused coordinates and in e gives a uniform bound. The determinant itself is the full principal minor, and $\det B \operatorname{tr} B^{-1}$ is the sum of the principal minors of order $k - 1$. This proves (5.7). As before, the uniform fiber bound also gives

$$(5.8) \quad \int_{\mathcal{T}} \det R \, ds \, d\omega \leq C.$$

Equations (5.7) and (5.8), together with Cauchy–Schwarz, give

$$\int_{\mathcal{T}} \sqrt{\det B \det R} \, ds \, d\omega \leq C.$$

For the inverse-Hessian weight, set

$$\theta = \frac{1}{2(1-\delta)} = \frac{n+2}{2(n+1)}, \quad t_B = \operatorname{tr} B^{-1}.$$

The interpolation identity is

$$(5.9) \quad \sqrt{\det B \det R} t_B = (I_0 t_B)^\theta (\det B t_B)^{1-\theta}.$$

Since $0 < \theta < 1$, Hölder's inequality applied to (5.9), using (5.6) and (5.7), gives

$$\int_{\mathcal{T}} \sqrt{\det B \det R} \operatorname{tr} B^{-1} \, ds \, d\omega \leq C.$$

The last two bounds control both terms in the target integral and hence $\sigma_j(Q)$. Its invariance under the preceding coordinate changes proves the lemma. $\square$

5.3. The growth contradiction.

Proof of Proposition 5.1. A model with one direction exists by Lemma 2.1. Suppose that the maximal number k is at least two, and fix the model and approximants used above. The case $k = n + 1$ has already been excluded in (5.3); hence $2 \leq k \leq n$.

Covering a logarithmic box by finitely many of the cells in Lemma 5.3 gives a constant C_{box} such that, for every fixed $L \geq 1$,

$$(5.10) \quad \sigma_j(Q_L) \leq C_{\text{box}}(L+1)^k \quad \text{for every sufficiently large } j.$$

The constant is independent of L and j . The index threshold may depend on L , because only finitely many cells are involved for each fixed L . By increasing that threshold, the tube coordinates are also available on Q_{L+2} .

For an integer $l \geq 1$, choose a smooth cutoff $\tilde{v}_l$ on $\mathbb{R}^k$, equal to one on $\{|t|_\infty \leq l\}$, zero outside $\{|t|_\infty < l+1\}$, and with a uniformly bounded Euclidean gradient. Its gradient is supported in the intervening shell. Proposition 4.1, applied with $v_l(s) = \tilde{v}_l(\log s)$, implies

$$\sigma_j(Q_l) \leq C_{\text{grow}}(\sigma_j(Q_{l+1}) - \sigma_j(Q_l))$$

whenever the tube is available through Q_{l+1} . Here C_{grow} is independent of l and j . Box boundaries have zero mass, because these measures have smooth densities in the tube coordinates. Writing $q = 1 + C_{\text{grow}}^{-1} > 1$ and taking an integer $l_0 \geq \max(1, L_0)$, iteration from (5.4) gives

$$(5.11) \quad \sigma_j(Q_L) \geq c_0 q^{L-l_0} \quad (L \geq l_0, L \text{ an integer})$$

for approximants on which all the indicated estimates hold.

Choose one finite integer L large enough that

$$c_0 q^{L-l_0} > C_{\text{box}}(L+1)^k.$$

Then choose one j large enough for the initial mass bound, the finitely many cell bounds through Q_L , and tube coordinates through Q_{L+2} . For this same j , (5.10) and (5.11) contradict each other. Thus no maximal model has $k \geq 2$, proving the proposition. $\square$

6. SECTION BALANCE AND RIGIDITY

We now convert the exclusion of models with two or more directions into uniform control of tangent sections. This control supplies the modulus of convexity needed for the interior estimates of Trudinger and Wang.

Lemma 6.1 (Uniform balance of tangent sections). *There is a constant $0 < \rho \leq 1$ such that, for every $a \in \Omega$ and $t > 0$, the compact convex body*

$$K_a(t) = \overline{S_u(a, t)} - a$$

satisfies

$$(6.1) \quad -\rho K_a(t) \subset K_a(t).$$

The same ρ works for all base points and heights.

Proof. For a fixed a , use the ambient affine coordinates

$$s = z - l_a(x), \quad y = x - a.$$

The epigraph lies in $s \geq 0$, contains the ray $\{s \geq 0, y = 0\}$, and has compact caps by Lemma 2.1. It is therefore a model with one direction, whose fiber at height t is $K_a(t)$. Proposition 5.1 says that the maximal direction number is one. The uniformity over models and coordinates in Proposition 2.6 consequently gives $-K_a(t) \subset CK_a(t)$ with one finite constant C . Taking $\rho = \min\{1, C^{-1}\}$ proves the claim. $\square$

Lemma 6.2 (Doubling and extension of rays). *The domain is $\Omega = \mathbb{R}^n$. Moreover, there are constants $\beta > 1$ and $A > 1$, depending only on ρ , such that every tangent-subtracted line restriction*

$$g(r) = u(a + re) - u(a) - r Du(a) \cdot e$$

satisfies

$$(6.2) \quad g(\beta r) \leq Ag(r) \quad (r > 0).$$

One may take

$$(6.3) \quad \beta = 1 + \frac{\rho}{2}, \quad A = \frac{(2 + \rho)(1 + \rho)}{\rho}.$$

Proof. Initially let $I = \{r : a + re \in \Omega\}$, an open interval containing zero, and choose $R \in I$ with $R > 0$. For $0 < r_0 < R$, the tangent section based at $a + r_0 e$ and having height

$$b = r_0 g'(r_0) - g(r_0) > 0$$

contains a on its boundary. Applying (6.1) to the displacement $-r_0 e$ puts the displacement $\rho r_0 e$ in the same translated section. In particular $(1 + \rho)r_0 \in I$, and the section inequality gives

$$g((1 + \rho)r_0) \leq (1 + \rho)r_0 g'(r_0).$$

Convexity and $g \geq 0$ give

$$g'(r_0) \leq \frac{g(R) - g(r_0)}{R - r_0} \leq \frac{g(R)}{R - r_0}.$$

Set

$$\theta = \frac{2 + \rho}{2(1 + \rho)}, \quad r_0 = \theta R.$$

Then $(1 + \rho)\theta = \beta$, with β as in (6.3), and

$$\beta R \in I, \quad g(\beta R) \leq \frac{\beta}{1 - \theta} g(R) = Ag(R).$$

Iterating the domain inclusion shows that the positive ray from a in direction e is contained in Ω . Every oriented direction admits an initial positive interval by openness, so all rays from a lie in Ω . This proves $\Omega = \mathbb{R}^n$, as well as (6.2) on every ray. $\square$

Fix a base point, translate it to zero, and subtract its tangent, so that $u(0) = 0$, $Du(0) = 0$, and $u \geq 0$. Each section $S_u(0, t)$ can be normalized by a linear map that preserves zero, with constants uniform in t . For $K = \overline{S_u(0, t)}$, apply Lemma 2.5 with $d = n$ and $C_* = \rho^{-1}$. It gives an invertible linear map L with

$$\overline{B_{2\rho/(1+\rho)}} \subset LK \subset \overline{B_{2n}}.$$

A scalar adjustment therefore produces invertible maps A_t and a fixed $R \geq 1$, depending only on n, ρ , for which

$$(6.4) \quad v_t(y) = t^{-1}u(A_t y), \quad B_1 \subset S_{v_t}(0, 1) \subset B_R.$$

The scalar adjustment may be made with a strict margin, so the inner inclusion holds for the open section. Balance and (6.2) are unchanged by these horizontal linear maps and positive vertical rescalings.

Lemma 6.3 (A uniform normalized modulus). *For all functions v_t in (6.4), there is one positive function $b : (0, \infty) \rightarrow (0, \infty)$ such that*

$$(6.5) \quad S_{v_t}(x, b(r)) \subset x + B_r \quad (x \in B_{1/4}, r > 0).$$

In particular, with $\gamma = (4R)^{-1}$,

$$(6.6) \quad S_u(x, tb(r)) \subset x + rS_u(0, t) \quad (x \in \gamma S_u(0, t), r, t > 0).$$

Proof. Write $v = v_t$. Since $v(0) = Dv(0) = 0$ and $v \leq 1$ on $\overline{B_1}$, convexity gives $0 \leq v(x) \leq 1/4$ on $B_{1/4}$. Fix such an x . For every unoriented line through x , choose its unit direction e so that $Dv(x) \cdot e \leq 0$. The ray in this direction meets $\partial S_v(0, 1)$ at a distance at most $2R$. At that intersection, its tangent-subtracted gap

$$g_{x,e}(r) = v(x + re) - v(x) - r Dv(x) \cdot e$$

is at least $1 - v(x) \geq 3/4$. Thus the positive-ray radius of the closed tangent section of height $1/4$, based at x , is at most $2R$. Balance of that section bounds the opposite-ray radius by $2R/\rho$: a point on the opposite ray, reflected and contracted by ρ , must still fit on the first ray. Consequently every unit direction e satisfies

$$(6.7) \quad g_{x,e}(R_*) \geq \frac{1}{4}, \quad R_* = \frac{2R}{\rho}.$$

The gap $g_{x,e}$ is nondecreasing on the positive ray. Define

$$(6.8) \quad N(r) = \max \left\{ 0, \left\lceil \log_\beta \frac{R_*}{r} \right\rceil \right\}, \quad b(r) = \frac{1}{8A^{N(r)}}.$$

Iterating (6.2) and using (6.7) gives

$$A^{N(r)} g_{x,e}(r) \geq g_{x,e}(\beta^{N(r)} r) \geq \frac{1}{4}.$$

Thus the tangent gap is at least $2b(r)$ at every point of distance r from x , and at every point farther away. This proves (6.5).

If $x \in \gamma S_v(0, 1)$, then $|x| < 1/4$. Since $B_1 \subset S_v(0, 1)$, we obtain

$$S_v(x, b(r)) \subset x + B_r \subset x + rS_v(0, 1).$$

Applying A_t gives (6.6). All constants and the function b are independent of the height t , the base x in the stated range, and the line direction. $\square$

Condition (6.6) is precisely the uniform strict convexity condition of [9, Condition (4.10)], with base point zero. For the final step we use its underlying interior estimate, so that the effect of the section rescaling is explicit.

Theorem 6.4 (Trudinger–Wang interior estimate). *Let $O \subset \mathbb{R}^n$ be a bounded convex domain and let $f \in C^4(O)$ have positive-definite Hessian and solve the affine maximal equation. Suppose $-1 \leq f \leq 0$. If a positive function b_0 gives the section control*

$$S_{f|_O}(x, b_0(r)) \subset B_r(x) \quad (x \in O, r > 0),$$

then $f \in C^\infty(O)$, and for each $O' \Subset O$ and integer $q \geq 2$ there are positive constants c, C_q such that

$$D^2 f \geq c \text{Id}, \quad |D^q f| \leq C_q \quad \text{on } O'.$$

The constants depend only on n , $\text{diam } O$, $\text{dist}(O', \partial O)$, and the specified lower modulus b_0 , with additional dependence on q for C_q .

This is [9, Theorem 4.2], stated in terms of a prescribed lower bound for the modulus of convexity. Its dimension is arbitrary. The sections $S_{f|_O}$ here are taken within O . Its proof combines the linearized Monge–Ampère estimates of Caffarelli and Gutiérrez [3], Caffarelli’s Monge–Ampère regularity theory [2], and Schauder estimates; see [9, Section 4]. The rescaling in the following proof is the reduction in [9, Theorem 2.1 and its proof], with the uniform modulus now supplied by Lemma 6.3.

Proof of Theorem 1.1. Proposition 5.1 applies for every $3 \leq n \leq 9$. Lemmas 6.1 and 6.2 therefore give $\Omega = \mathbb{R}^n$ and uniform section balance. Fix any base point, translate it to zero, subtract its tangent, and use the normalized functions v_t from (6.4).

These functions solve the same affine maximal equation. Indeed, for $v(y) = t^{-1}u(Ay)$, set

$$c_t = t^{-n}(\det A)^2, \quad a = \frac{n+1}{n+2}.$$

Then

$$w_v = c_t^{-a} w_u \circ A, \quad U_v = t^{1-n}(\det A)^2 A^{-1}(U_u \circ A)A^{-\top},$$

so the contraction $U_v^{ij}(w_v)_{ij}$ is a nonzero constant multiple of $(U_u^{ij}(w_u)_{ij}) \circ A$, and vanishes. Subtracting an affine function leaves the equation unchanged as well.

Apply Theorem 6.4 to $f_t = v_t - 1$ on $O = B_{1/4}$, with $O' = B_{1/8}$. There $-1 \leq f_t \leq -3/4$, and Lemma 6.3 supplies the same lower modulus at every point of O . Restricting a section to O only decreases it. We obtain constants independent of t such that

$$(6.9) \quad D^2 v_t(0) \geq c_0 \text{Id}, \quad |D^3 v_t(0)| \leq C_0.$$

Put $H = D^2 u(0) > 0$. The identity

$$D^2 v_t(0) = t^{-1} A_t^\top H A_t$$

and the first bound in (6.9) imply

$$tc_0 |z|^2 \leq (A_t z)^\top H (A_t z) \leq \lambda_{\max}(H) |A_t z|^2,$$

and hence

$$(6.10) \quad \|A_t^{-1}\| \leq \sqrt{\lambda_{\max}(H)/c_0} t^{-1/2}.$$

For any three vectors z_1, z_2, z_3 , differentiation of $u(x) = t v_t(A_t^{-1}x)$ gives

$$D^3 u(0)[z_1, z_2, z_3] = t D^3 v_t(0)[A_t^{-1} z_1, A_t^{-1} z_2, A_t^{-1} z_3].$$

Equations (6.9) and (6.10) therefore give

$$|D^3 u(0)| \leq C t^{-1/2} \longrightarrow 0 \quad \text{as } t \longrightarrow \infty.$$

The base point was arbitrary, so $D^3u \equiv 0$. Thus

$$u(x) = \frac{1}{2}x^T Qx + b \cdot x + c$$

on $\mathbb{R}^n$, with $Q > 0$. Its graph is an invertible affine image of $\{(z, |z|^2) : z \in \mathbb{R}^n\}$, as required. $\square$

The affine-complete consequence. The graph theorem now gives the consequence announced in the introduction. The completeness and global graph results of Trudinger and Wang represent the hypersurface as a complete convex graph.

Corollary 6.5 (Affine-complete hypersurfaces). *Let $3 \leq n \leq 9$. Let M be a smooth locally uniformly convex immersed hypersurface in $\mathbb{R}^{n+1}$ whose underlying manifold is connected and open (noncompact and without boundary). Suppose that M is complete for the affine (Berwald–Blaschke) metric and is classically affine maximal (stationary for affine area under compactly supported smooth variations). Then M is an invertible affine image of the standard elliptic paraboloid.*

Proof. By [10, Theorem A], M is complete for the induced Euclidean metric. By [10, Corollary 1], an invertible affine change of coordinates gives an affine image $\widetilde{M}$ that is the global graph of a convex function u over a nonempty open convex domain $\Omega \subset \mathbb{R}^n$, not assumed to equal $\mathbb{R}^n$. The linear part L of this change uniformly compares Euclidean lengths by its least and greatest singular values, preserving completeness. By Equation (3.2), it multiplies affine area on compact patches by the fixed positive factor $|\det L|^{n/(n+2)}$, preserving stationarity. Smoothness and local uniform convexity give $u \in C^\infty(\Omega)$ with positive-definite Hessian, and classical stationarity gives (1.1). Theorem 1.1 now implies $\Omega = \mathbb{R}^n$ and that u is a quadratic polynomial with positive-definite Hessian. Undoing the affine change of coordinates proves the conclusion. $\square$

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